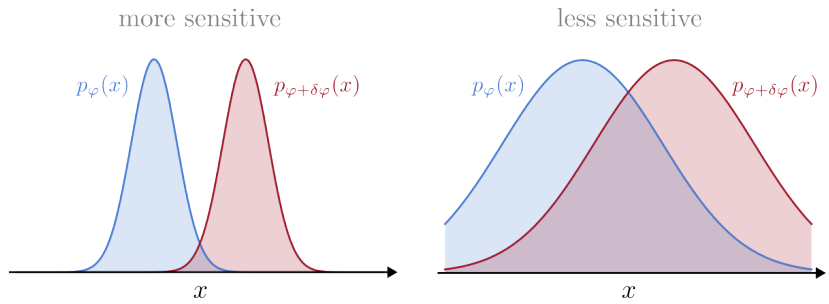

Figure 4.1: Parametrised probability distributions with different sensitivities to parameter changes. *Left:* A familiy with high sensitivity. A small parameter change $\delta\varphi$ results in distributions with little overlap that can be well distinguished. *Right:* A familiy with low sensitivity. The same parameter change results in distributions with large overlap that are difficult to distinguish.

4.1.1 Fisher information

How can we tell how good a given measurement procedure is for estimating some parameter? Or, in different terms, how much information the measurement data contains about the parameter?

In general, for a fixed value of the parameter, the measurement will not provide a deterministic outcome but rather a range of outcomes will occur with different probabilities. When the parameter value changes, the probability distribution over the outcomes will also change. Intuitively, the more the distribution changes for a small change in the parameter, the more sensitive the measurement is and the better it will be for estimation. The (classical) *Fisher information* makes this intuition precise.

The Fisher information quantifies the amount of information about a parameter φ carried by a random variable X with probability distribution $p_\varphi(x)$. It is defined by

$$\mathcal{F}_\varphi(X) = \sum_x \left[\frac{\partial}{\partial\varphi}\log(p_\varphi(x))\right]^2 p_\varphi(x). \tag{4.1}$$

If X is continuous, the sum over x is replaced by an integral and $p_\varphi(x)$ should be understood as the probability density. Why does this measure sensitivity? The quantity inside the bracket can be written

$$\frac{\partial}{\partial\varphi}\log(p_\varphi(x)) = \frac{1}{p_\varphi(x)}\frac{\partial p_\varphi(x)}{\partial\varphi} = \frac{\dot{p}_\varphi(x)}{p_\varphi(x)}, \tag{4.2}$$

where the dot denotes the derivative $\frac{\partial}{\partial\varphi}$. This quantity is thus the relative change in the distribution p_φ at the point x corresponding to a small change in φ. It is also know as the *score*. Because the distribution is normalised, the mean value of the score averaged over all x vanishes (increases in the probabilities for some x must be compensated by decreases for others to keep the total change zero – see Prob. 4.1). It follows that the Fisher information is the variance of the score, i.e. the variance of the relative change in probability when the parameter is shifted slightly. A high Fisher information implies that p_φ changes a lot when φ changes, while a low Fisher information implies that

φ has little influence on p_φ. This is illustrated in Fig. 4.1. Note that $\mathcal{F}_\varphi \geq 0$ always.

Example 4.1: Fisher information of measurements with Gaussian noise

Consider a measurement of some quantity φ subject to Gaussian noise, such that the distribution of the measurement outcome X is given by

$$p_\varphi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\varphi)^2}{2\sigma^2}\right). \tag{4.3}$$

That is, the average of the outcomes equal the true value of the parameter, $\bar{X} = \int x p_\varphi(x) dx = \varphi$ but they fluctuate around this value following a Gaussian with variance σ^2.

How sensitive is this measurement to φ? Intuitively, the stronger the noise – i.e. the larger the variance – the less sensitive the measurement is. The Fisher information captures this. The Fisher information corresponding to (4.3) is

$$\mathcal{F}_\varphi(X) = \int_{-\infty}^{\infty} \left[\frac{\partial}{\partial\varphi}\left(-\frac{(x-\varphi)^2}{2\sigma^2} - \frac{1}{2}\log\left(2\pi\sigma^2\right)\right)\right]^2 p_\varphi(x) dx \tag{4.4}$$

$$= \frac{1}{\sigma^4}\int_{-\infty}^{\infty} (x-\varphi)^2 p_\varphi(x) dx = \frac{1}{\sigma^2}. \tag{4.5}$$

Thus, in this case, $\mathcal{F}_\varphi(X)$ is just the inverse of the variance. When the distribution of measurement outcomes broadens, sensitivity decreases as expected. Note also that in the limit of vanishing noise $\mathcal{F}_\varphi(X)$ diverges. This reflects that a noiseless, perfect measurement which always returns the true value of the parameter can distinguish arbitrarily small changes. Of course, such measurements cannot be realised in practice.

4.1.2 Estimators

While the Fisher information is defined in terms of the probability distribution of measurement outcomes, in any real experiment there will be access only to a finite amount of data sampled from this distribution. The parameter of interest must be estimated from this finite sample. The Fisher is useful because it allows us to bound the precision of such estimates (through the Cramér-Rao bound, as discussed below).

An *estimator* is a map from sample data to the domain of the parameter being estimated. It corresponds to our best guess for the parameter value given the observations. Suppose a parameter φ is to be estimated from a sample of N measurement outcomes corresponding to a joint, random variable X. For a sample $\boldsymbol{x} = (x_1, \ldots, x_N)$, we denote the estimate by $\tilde{\varphi}(\boldsymbol{x})$. That is, the true parameter value is φ and the estimate is denoted by a tilde[1].

[1] In literature on estimation, it is common to see the notation $\hat{\varphi}$. We do not adopt that here to avoid confusion with operators in quantum mechanics.